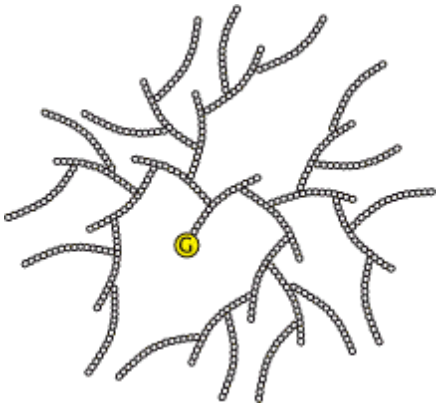
Glycogen Metabolism



PART-1: GLYCOGEN SYNTHESIS

LEARNING OBJECTIVES

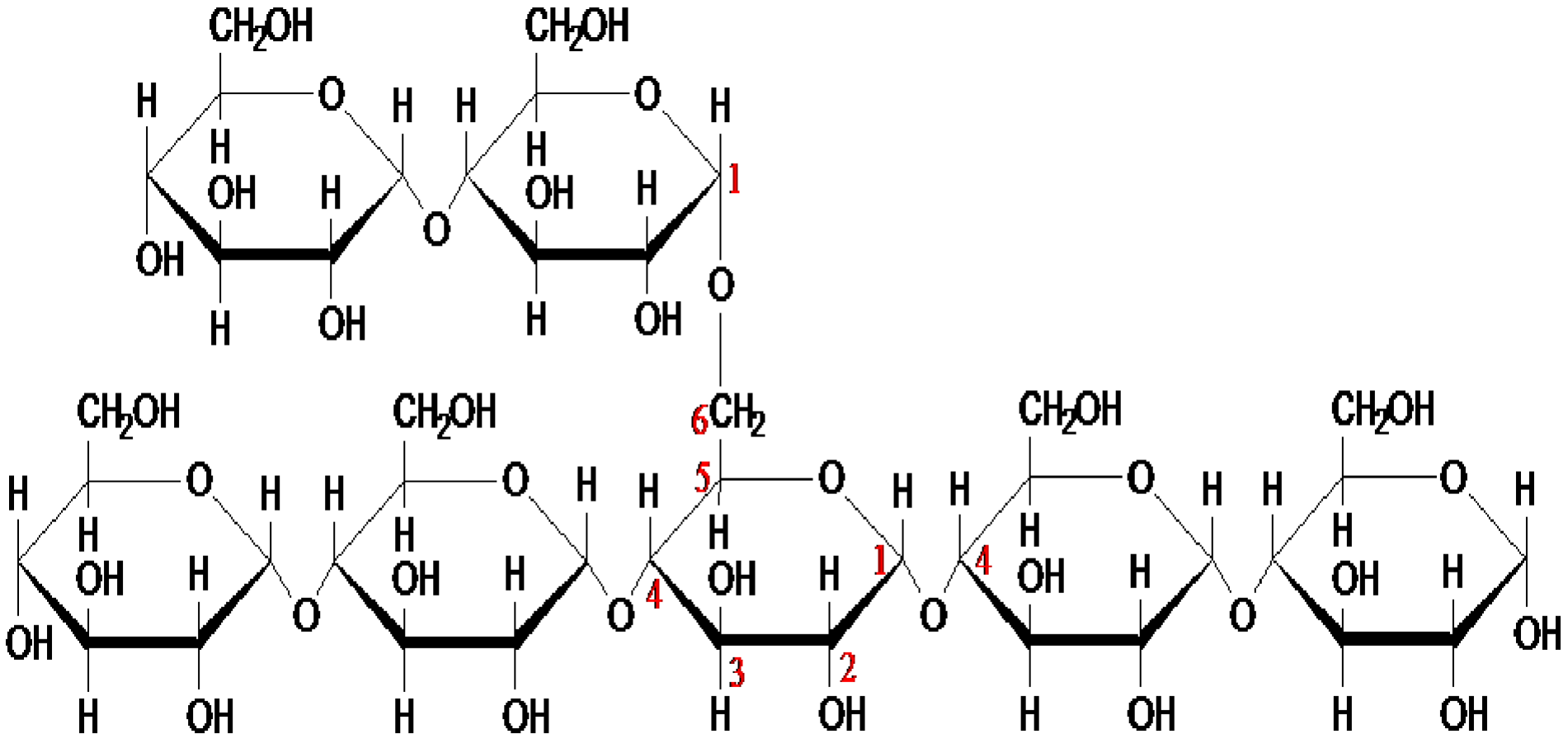
At the end of the lecture, students should be able to:

- Define glycogenesis
- Describe the structure and function of glycogenin
- Enumerate steps of glycogenesis with details
- Discuss allosteric regulation and covalent modifications under the influence of hormones.

Glycogen– Introduction

- Glycogen is a **readily mobilized storage form of glucose.**
- It is a very large, branched polymer of glucose residues that can be broken down to yield glucose molecules when energy is needed.
- Most of the glucose residues in glycogen are linked by **α-1,4-glycosidic bonds**.
- **Branches at about every tenth residue are created by α-1,6-glycosidic bonds**.

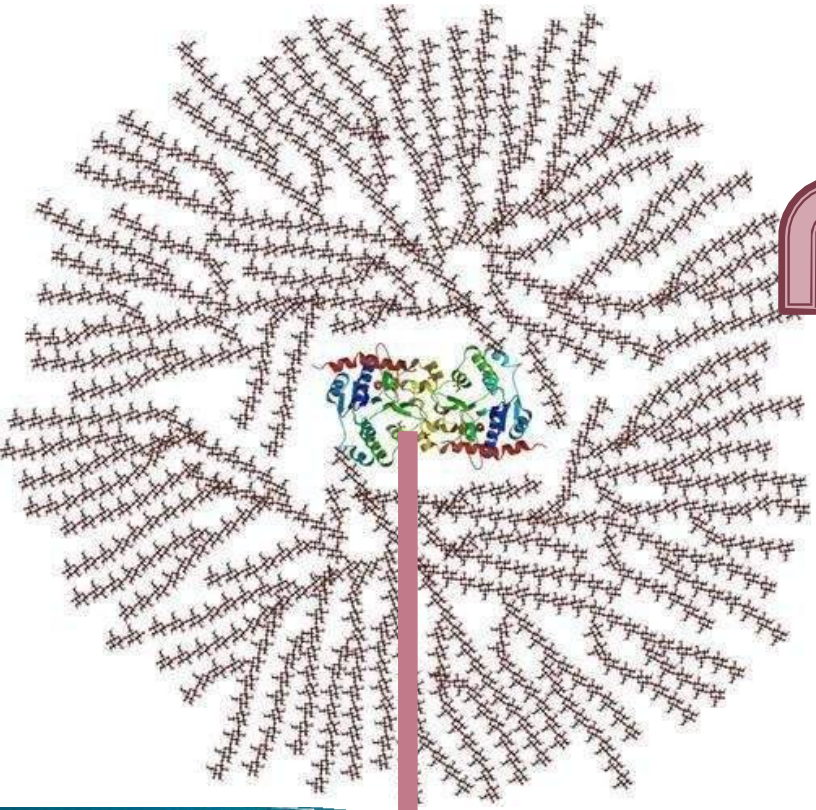
Glycogen-Chemistry



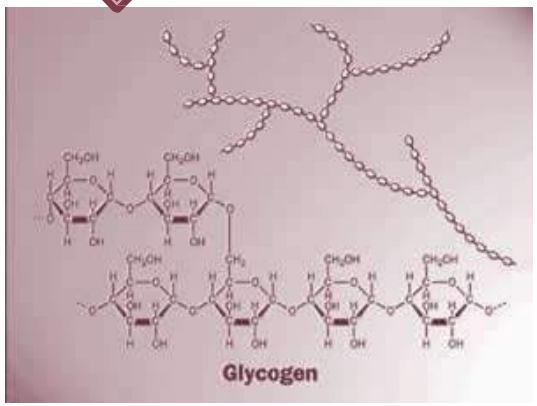
Glycogen–Chemistry

- Glycogen is present in the cytosol in the form of granules ranging **in diameter from 10-40 nm.**
- It has a **molecular mass of 10^7 Da** and consists of polysaccharide chains, each containing about 13 glucose residues.
- The chains are either branched or unbranched and are arranged in 12 concentric layers.
- The **branched chains (each has two branches) are found in the inner layers** and the unbranched chains in the outer layer. (G, Glycogenin, the primer molecule for glycogen synthesis.)

Glycogen– Chemistry



Glycogenin



The highly branched structure of glycogen provides a large number of sites for glycogenolysis, permitting rapid release of glucose 1-phosphate for muscle activity.

Glycogen Storage Sites

- It is stored mainly in **liver and muscle**
- The liver content of glycogen is greater than that of muscle,
- Since the muscle mass of the body is considerably greater than that of the liver, about **three-quarters of total body glycogen is in muscle**

Storage of Carbohydrate in a 70-Kg Human

	Percentage of Tissue Weight	Tissue Weight	Body Content
Liver glycogen	5.0	1.8 kg	90 g
Muscle glycogen	0.7	35 kg	245 g
Extracellular glucose	0.1	10 L	10 g

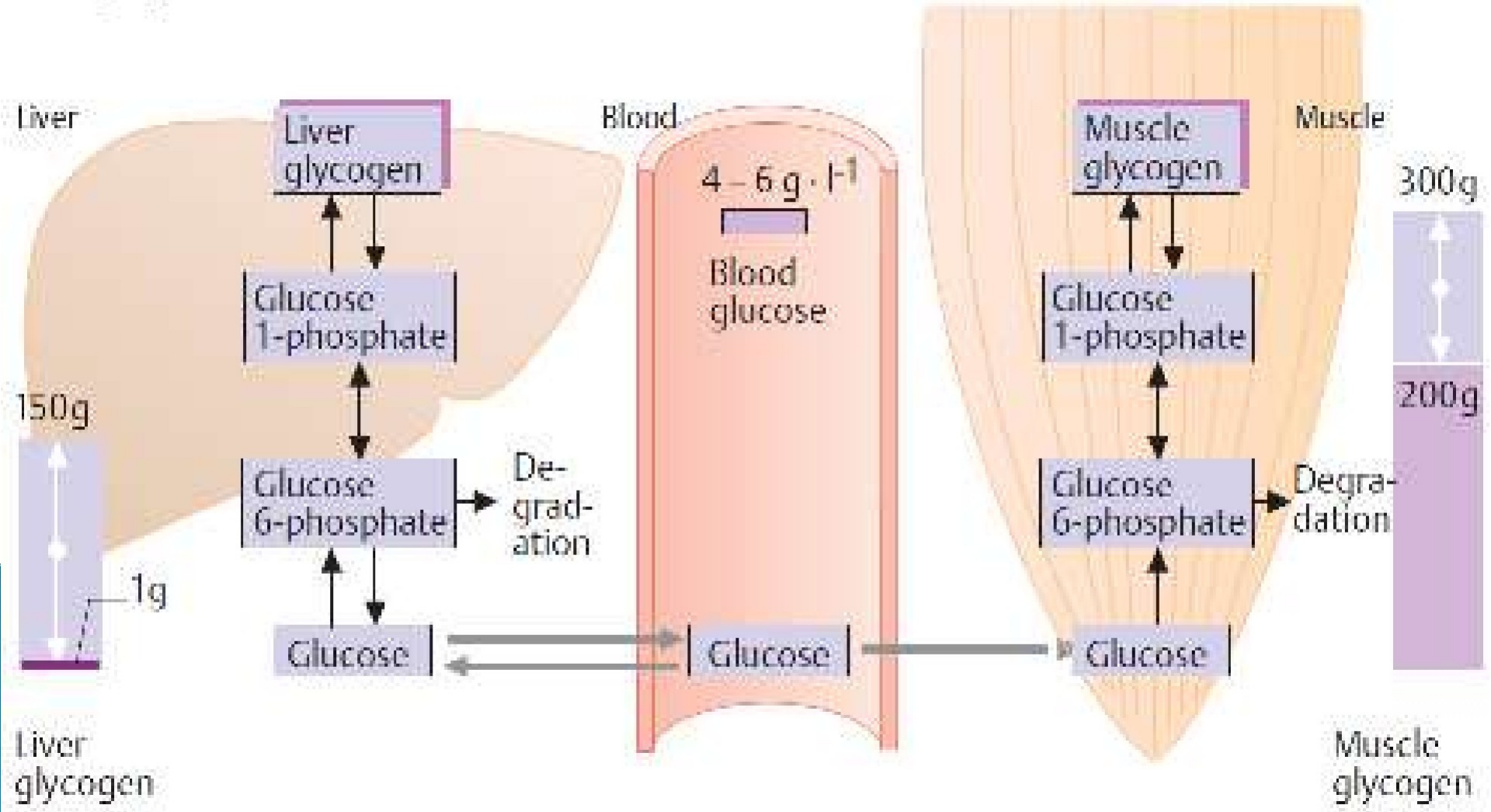
Reasons for storing Glycogen as a fuel–

- Glycogen serves as a **buffer** to maintain blood glucose levels.
- Glucose is virtually the **only fuel used by the brain,** except during prolonged starvation.
- The glucose from glycogen is **readily mobilized and is therefore a good source of energy for sudden, strenuous activity**.
- Unlike fatty acids, the released glucose can **provide energy in the absence of oxygen** and can **thus supply energy for anaerobic activity.**

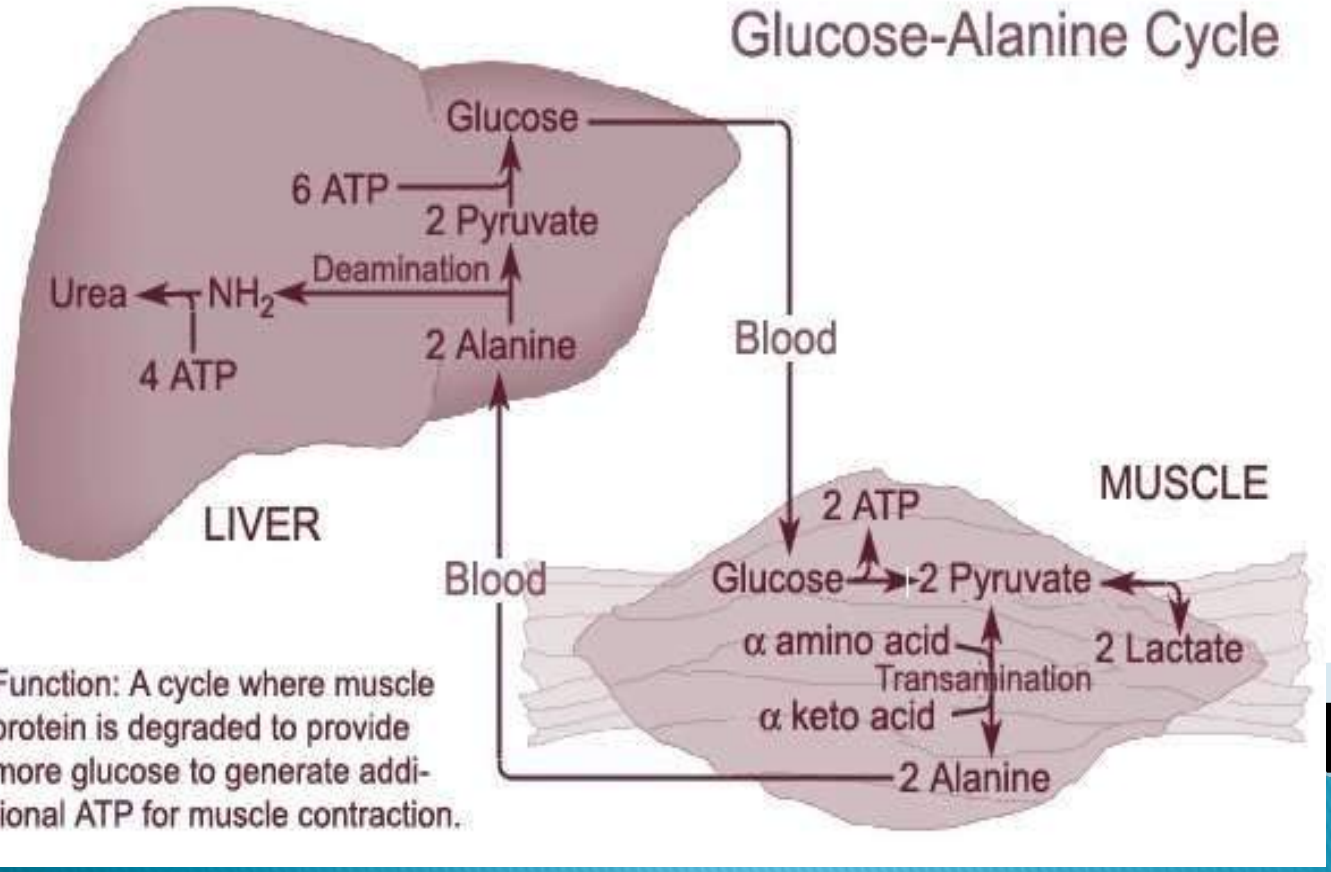
Glycogenesis

- **Glycogenesis is the synthesis of glycogen from glucose.**
- Glycogenesis mainly occurs in **muscle and liver.**
- Muscle glycogen provides a **readily available source of glucose for glycolysis** within the muscle itself.
- **Liver glycogen functions to store and export glucose to maintain blood glucose between meals.**

Liver and Muscle glycogen



Indirect contribution by muscle to maintain blood glucose levels



- Alanine and lactate transported from muscle are used for glucose production in liver by gluconeogenesis.
- Glucose is poured in blood to maintain homeostasis.

Steps of Glycogenesis

1) **Activation of Glucose**
2) **Initiation**
3) **Elongation**
4) **Glycogen branching**

Activation of Glucose

- Synthesis of glycogen from glucose is carried out by the enzyme **glycogen synthase.**
- This enzyme utilizes **UDP-glucose** as one substrate and the non-reducing end of glycogen as another.
- **UDP-glucose, the glucose donor in the biosynthesis of glycogen, is an activated form of glucose.**

Activation of Glucose (Contd.)

UDP-glucose is formed from glucose-1-phosphate:

$$\text{Glucose 1-phosphate} + \text{UTP} \rightleftharpoons \text{UDP-glucose} + PP_i$$

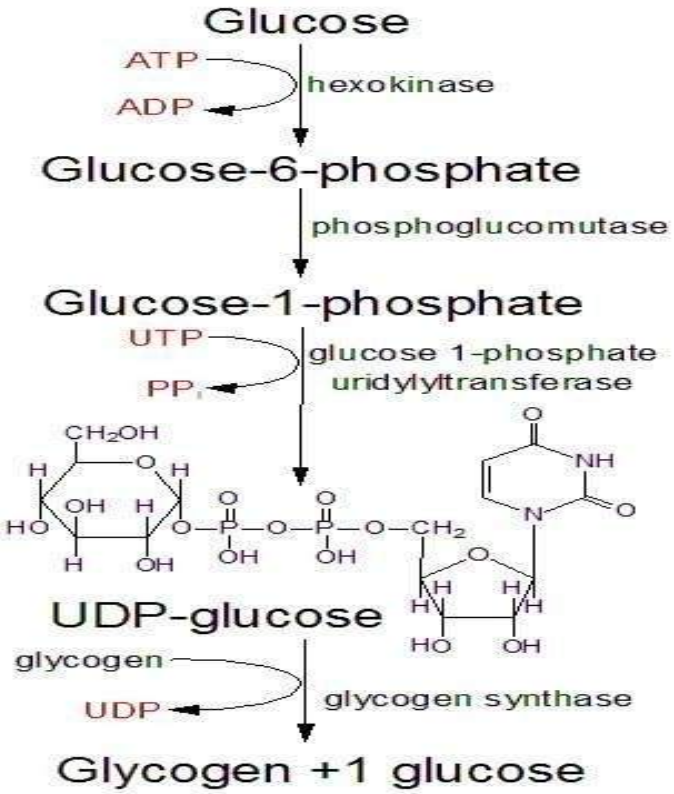
$$PP_i + H_2O \longrightarrow 2\,P_i$$

$$\text{Glucose 1-phosphate} + \text{UTP} + H_2O \longrightarrow \text{UDP-glucose} + 2\,P_i$$

Spontaneous hydrolysis of the ~P bond in PP_i (P~P) drives the overall reaction.

Cleavage of PP_i is the only energy cost for glycogen synthesis (<u>one</u> ~P bond per glucose residue).

Activation of Glucose (Contd.)

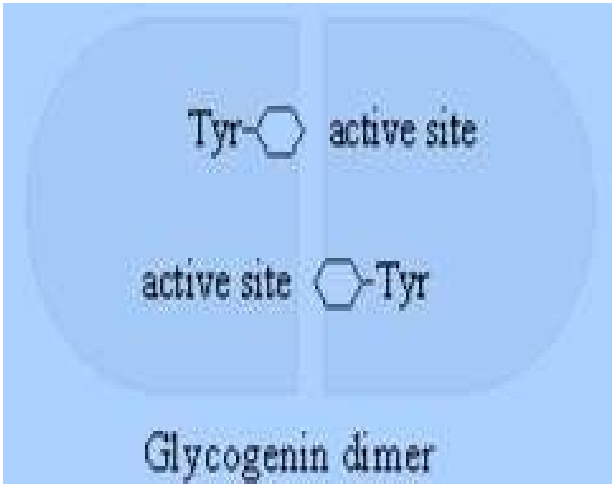


As in glycolysis, glucose is phosphorylated to glucose 6-phosphate, catalyzed by **hexokinase** in muscle and **glucokinase** in liver. Glucose 6-phosphate is isomerized to glucose 1-phosphate by **Phosphoglucomutase.**

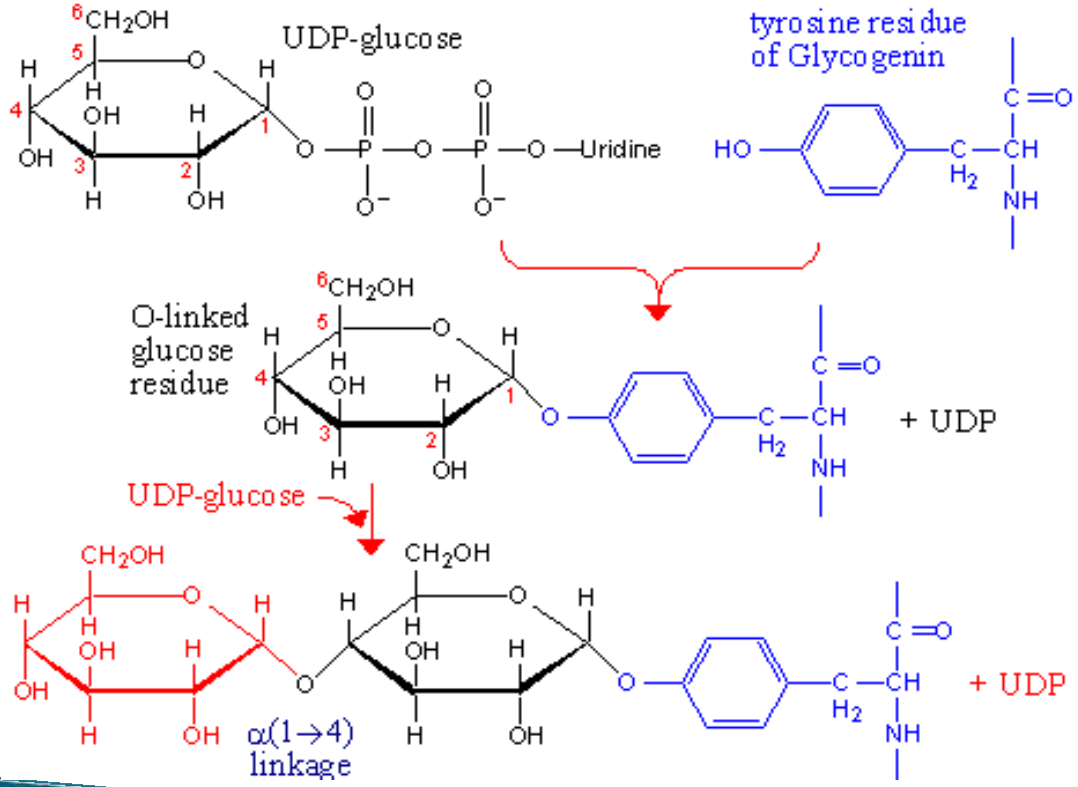
Glucose 1-phosphate reacts with uridine triphosphate (UTP) to form the active nucleotide **uridine diphosphate glucose (UDPGlc)** and pyrophosphate. The reaction is catalyzed by **UDPGlc pyro phosphorylase.**

2) Initiation

- Glycogen synthase can add glucosyl residues only if the polysaccharide chain already contains more than four residues.
- Thus, glycogen synthesis requires **a primer**.
- This priming function is carried out by **glycogenin,**
- A protein composed of two identical 37-kd subunits, each bearing an oligosaccharide of alpha-1,4-glucose units.



Initiation (Contd.)



A **glycosidic bond** is formed between the anomeric C1 of the glucose moiety derived from UDP-glucose and the hydroxyl oxygen of a **tyrosine** side-chain of **Glycogenin**.

UDP is released as a product.

Each subunit of glycogenin catalyzes the addition of eight glucose units to its partner in the glycogenin dimer. At this point, glycogen synthase takes over to extend the glycogen molecule.

3) Elongation

- New glucosyl units are added to the **nonreducing terminal** residues of glycogen.

- The activated glucosyl unit of UDP glucose is transferred to the hydroxyl group at a C-4 terminus of glycogen to form **an α-1,4-glycosidic linkage**.

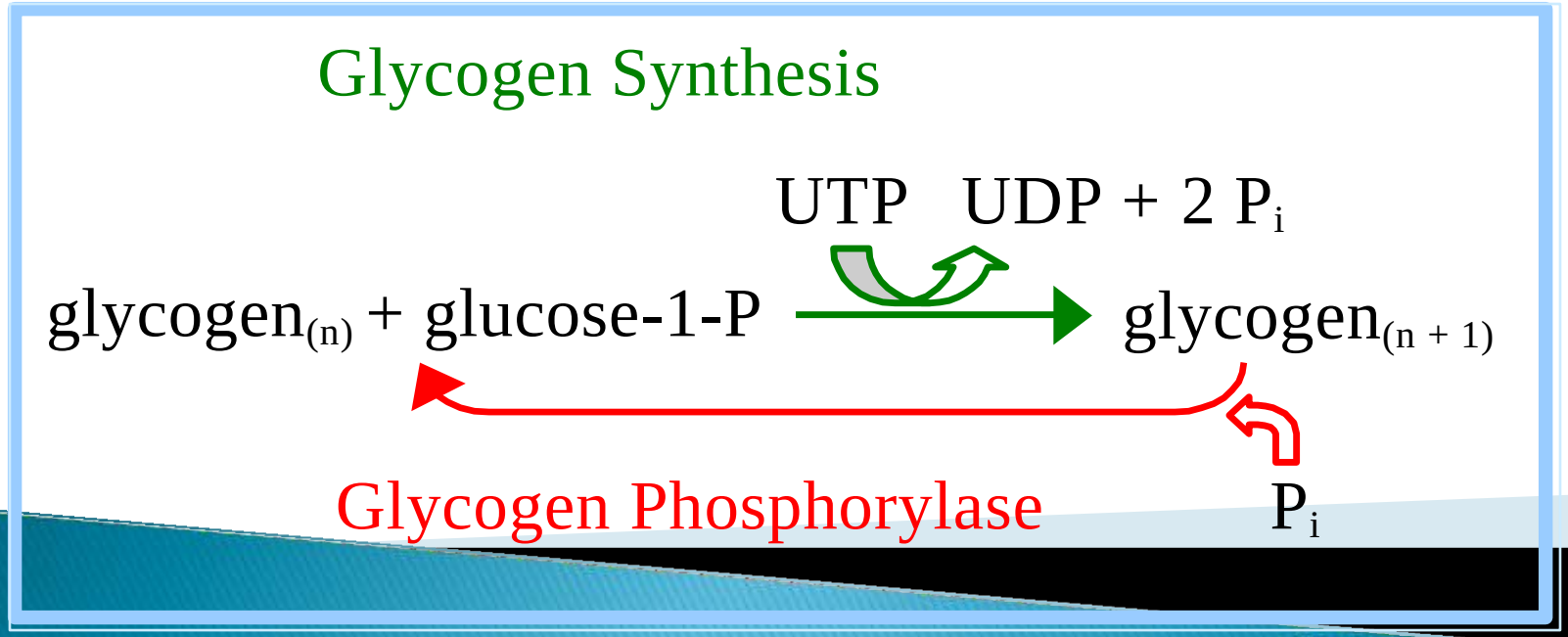
- In elongation, UDP is displaced by the terminal hydroxyl group of the growing glycogen molecule.

- This reaction is catalyzed by **glycogen synthase, the key regulatory enzyme in glycogen synthesis**.

3) Elongation

Both synthesis & breakdown of glycogen are spontaneous.

If both pathways were active simultaneously in a cell, there would be a "**futile cycle**" with cleavage of **one ~P bond per cycle** (in forming UDP-glucose).To prevent this both pathways are reciprocally regulated



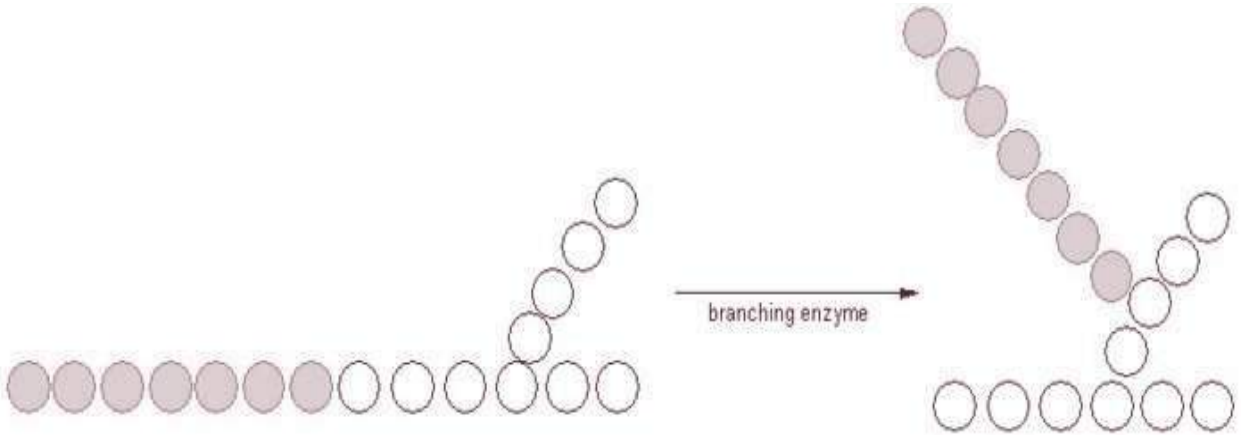
4) Glycogen branching

- Glycogen synthase catalyzes only the synthesis of α-1,4 linkages.
- Another enzyme is required to form the α-1,6 linkages that make glycogen a branched polymer.
- Branching occurs after a number of glucosyl residues are joined in α-1,4 linkage by glycogen synthase.
- A branch is created by the breaking of an α-1,4 link and the formation of an α-1,6 link.

4) Glycogen branching(Contd.)

- A block of residues, typically 7 in number, is transferred to a more interior site.
- The **branching enzyme** that catalyzes this reaction is quite exacting.
- The **block of 7** or so residues must include the nonreducing terminus and come from a chain at least 11 residues long.
- In addition, **the new branch point must be at least 4 residues away from a preexisting one**.

4) Glycogen branching(Contd.)



Branching is important *because* **it increases the solubility of glycogen**. Furthermore, branching creates a large number of terminal residues, the **sites of action of glycogen phosphorylase and synthase**. Thus, **branching increases the rate of glycogen synthesis and degradation.**

Energy Expenditure for Glycogenesis

Glucose 6-phosphate $\longrightarrow$ glucose 1-phosphate (1)

Glucose 1-phosphate + UTP $\longrightarrow$ UDP-glucose + PP_i (2)

$PP_i + H_2O \longrightarrow 2\,P_i$ (3)

UDP-glucose + $glycogen_n \longrightarrow glycogen_{n+1}$ + UDP (4)

UDP + ATP $\longrightarrow$ UTP + ADP (5)

Sum: Glucose 6-phosphate + ATP + $glycogen_n + H_2O \longrightarrow$ $glycogen_{n+1}$ + ADP + $2\,P_i$

Regulation of Glycogen Metabolism

The principal enzymes controlling glycogen metabolism—**glycogen phosphorylase and glycogen synthase**—are regulated by **allosteric mechanisms and covalent modifications** due to reversible phosphorylation and dephosphorylation of enzyme protein in response to hormone action

Regulation of Glycogenesis(Allosteric modification)

- **Glycogen Synthase** is allosterically **activated** by **glucose-6-P.**
- Thus Glycogen Synthase is active when high blood glucose leads to elevated intracellular glucose-6-P.
- It is useful to a cell to store glucose as glycogen when the input to Glycolysis (glucose-6-P), and the main product of Glycolysis (ATP), are adequate.

Regulation of Glycogenesis by covalent modification (phosphorylation):

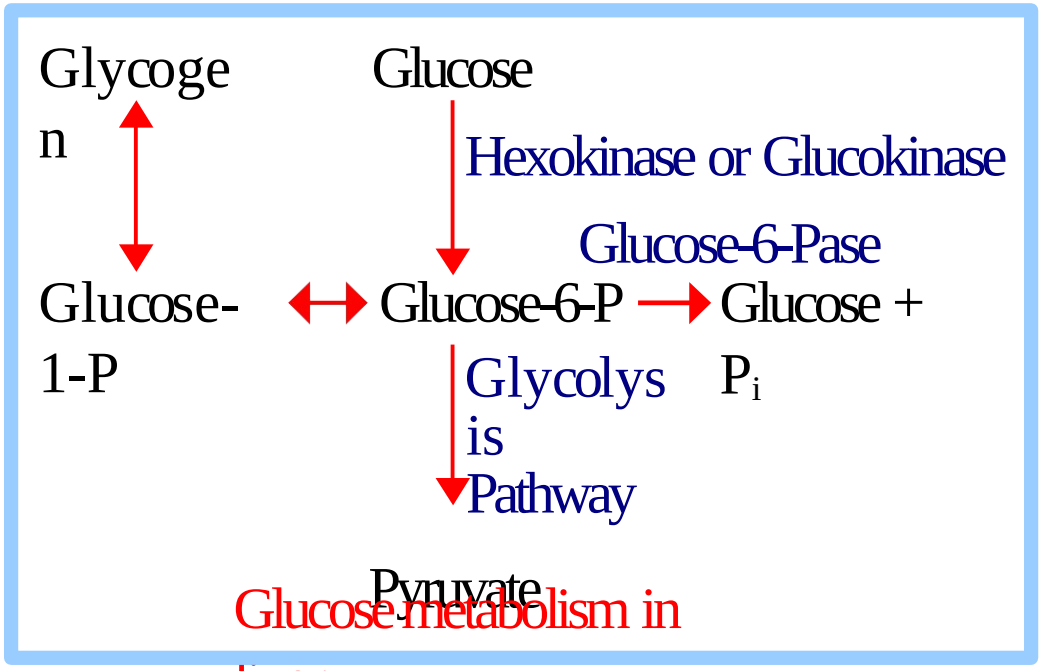
- The hormones **glucagon** and **epinephrine** activate G-protein coupled receptors to trigger **cAMP cascades**.
- Both hormones are produced in response to **low blood sugar**.
- Glucagon, which is synthesized by α -cells of the pancreas, activates cAMP formation in liver.
- Muscle cells lack Glucagon receptors
- Epinephrine activates cAMP formation in muscle

Regulation of Glycogenesis by covalent modification (phosphorylation):

- Glycogen synthase exists in both phosphorylated or nonphosphorylated states
- **Active glycogen synthase a** is dephosphorylated and **inactive glycogen synthase b** is phosphorylated
- The cAMP cascade results in **phosphorylation** of a serine hydroxyl of Glycogen synthase, which **promotes transition to the inactive state.**

Regulation of Glycogenesis by covalent modification (phosphorylation):

- **Phosphorylation** of Glycogen Synthase promotes the **"b"** (less active) conformation.
- The cAMP cascade thus **inhibits glycogen synthesis.**
- Instead of being converted to glycogen, glucose-1-P in liver may be converted to glucose-6-P, and dephosphorylated for release to the blood.



Glucose metabolism in liver.

Role of Insulin In Glycogenesis

- **Insulin**, produced in response to **high blood glucose**, triggers a separate signal cascade that leads to **activation of Phosphoprotein Phosphatase**.
- This phosphatase catalyzes removal of regulatory phosphate residues from Glycogen Synthase enzyme.
- Thus **insulin antagonizes** effects of the cAMP cascade induced by **glucagon** & **epinephrine**.
- cAMP is hydrolyzed by **phosphodiesterase**, so terminating hormone action; in liver insulin increases the activity of phosphodiesterase.

REFERENCES

- Harper
- Lippincott